LINUX SYSTEM ADMIN PLAYBOOK

Real-World Interview Questions & Answers

For Experienced Professionals

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Linux System Admin – Interview Questions & Answers

Q1. How to set password for a user that never expire ?

Use below command to disable password expiry:

chage -M 99999 username

or

passwd -x -1 username

This ensures the password will never expire.

Q2. Why /etc/passwd and /etc/shadow file cannot be merged ?

Because of security reasons.
/etc/passwd file is readable by all users in the system, while /etc/shadow contains encrypted passwords and is accessible only by root. If both are merged, sensitive password data will be exposed to all users.

Q3. How we can list all files opened by particular process?

Use:

lsof -p <PID>

This will list all open files by that process.

Q4. We are unable to unmount a filesystem. What can be the possible reason ?

Possible reasons:

- Filesystem is in use by some process
- Current directory is inside that mount point
- Open files or running applications using it

Check using:

lsof | grep <mount_point>
fuser -m <mount_point>

Q5. What can be the reason if a server is taking more time after reboot?

- Too many services enabled at startup
- Filesystem check (fsck) running
- Network/DNS delay
- Hardware issues
- High resource usage

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Q6. We are not able to create new files in a partition. We have enough disk space and permissions. What can be the possible reason ?

The partition may have run out of inodes.
Check using:

```
df -i
```

Q7. How to check kernel routing information?

Use:

```
route -n
```

or

```
ip route
```

Q8. How to set sticky bit and different b/w small 's' and capital 'S'?

Set sticky bit:

```
chmod +t directory
```

Difference:

- small s → execute permission is present
- capital S → execute permission is missing

Q9. which file is used to specify default gateway ?

In older systems:

/etc/sysconfig/network

In modern systems, default gateway is configured using:

```
ip route
```

Q10. How to switch between different run-levels ?

Old method:

```
init 3
init 5
```

Systemd:

```
systemctl isolate multi-user.target
```

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Q11. What is NFS ? How to configure and what port is uses?

NFS (Network File System) is used to share files over network.
Default port: **2049**

Basic steps:

- Configure /etc/exports
- Run:

exportfs -a

- Mount:

mount -t nfs server:/share /mnt

Q12. What is nice and renice value. How to set them for a particular process.

Nice value defines process priority.

- Range: -20 (high priority) to 19 (low priority)

Set nice:

nice -n 10 command

Change priority:

renice 5 -p PID

Q13. Difference between FTP and TFTP?

- FTP uses TCP, supports authentication and is secure
- TFTP uses UDP, no authentication, used for simple transfers

Q14. How to extend logical volume size in Linux ?

lvextend -L +5G /dev/vg/lv
resize2fs /dev/vg/lv

Q15. How will you rollback packages after patching ?

yum history undo <ID>

Q16. What is rsync? tell us the command and also how it is different from scp ?

Rsync is used for efficient file synchronization.

rsync -avz source/ destination/

Difference:

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- rsync copies only changed data
- scp copies full data every time

Q17. How to reduce the size of logical volume ? Filesystem reduction can be happened online or not ?

Steps:

umount
e2fsck
resize2fs
lvreduce

Filesystem reduction **cannot be done online**, it must be unmounted.

Q18. How to set line no. in vim ?

:set number

Q19. Define each field of /etc/fstab file ?

1. Device name
2. Mount point
3. Filesystem type
4. Mount options
5. Dump
6. Pass (fsck order)

Q20. Difference between soft link and hard link ? Can we create a hard link across filesystems?

- Soft link → symbolic link, can cross filesystem
- Hard link → same inode, cannot cross filesystem

Q21. How to check a port is listening or not ?

netstat -tulnp

or

ss -tulnp

Q22. Difference between df and du command?

- df → shows filesystem disk usage
- du → shows file/directory usage

Q23. What is journaling ? Difference between ext2 and ext fs?

Journaling logs changes before writing.

- ext2 → no journaling
- ext3/ext4 → journaling available

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Q24. Define boot process step by step ?

1. BIOS/UEFI
2. Bootloader (GRUB)
3. Kernel loading
4. init/systemd
5. Services start

Q25. What are Run-levels ?

Runlevels define system state:

- 0 → shutdown
- 1 → single user
- 3 → multi-user
- 5 → GUI
- 6 → reboot

Q26. what is meaning of PCPU and JCPU in w command output?

- PCPU → CPU usage by process
- JCPU → total CPU time of job

Q27. What is load average ? and how it is calculated ?

It shows average system load over:

- 1 minute
- 5 minutes
- 15 minutes

Calculated based on runnable processes and CPU cores.

Q28. What happen when you type a url in browser and hit enter ?

- DNS lookup
- TCP connection
- HTTP request
- Server response
- Browser rendering

Q29. Diff between RPM and YUM? How to check package is installed or not?

- RPM → basic package install
- YUM → dependency handling

Check:

rpm -qa | grep package

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Q30. What is ACL? Why we need it and how to check or set ACL over a file or directory ?

ACL provides additional permissions.

Check:

getfacl file

Set:

setfacl -m u:user:rwx file

Q31. What is inode and what things it stored ?

Inode stores:

- permissions
- owner
- size
- timestamps
- block location

Q32. Can we schedule a 2 sec task in crontab ?

No, cron supports minimum 1 minute.
Use script with sleep.

Q33. In a production env if a server rebooted itself, how would you check ?

last -x
who -b

Check logs:

/var/log/messages
/var/log/secure

Q34. Which command is used to set user password expiry date ?

chage -E YYYY-MM-DD username

Q35. How to extract a single file from tar archive ?

tar -xvf file.tar filename

Q36. Command to check memory usage ?

free -h
top

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Q37. Command to check kernel version and architecture ?

uname -r
uname -m

Q38. How to search text within multiple files ?

grep -r "text" *

Q39. what is atime, mtime and ctime? How we can take advantage of them ?

- atime → last access
- mtime → last modification
- ctime → metadata change

Used for monitoring and cleanup.

Q40. How to delete log files which are 30 days old ?

find /path -type f -mtime +30 -delete

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